

Practice 1: The coordinate plane

Name: _____ Date: _____

Draw grids in the work area with equally spaced units from 0 to 10 on both axes unless another range is given. Graph paper is optional. Label points and show your reasoning.

1. In an ordered pair, what does the first number tell you? What does the second number tell you? Explain how you would plot $(4, 2)$.

Work space

2. Draw a coordinate plane with axes from 0 to 6. Plot $A(2, 5)$ and $B(5, 1)$ and label them.

Work space

Practice 2: Reading coordinates

Name: _____ Date: _____

1. Give the ordered pair for a point 3 units right and 6 units up from the origin.

Work space

2. Point C is at $(7, 4)$. State its x-coordinate and y-coordinate, then describe how to reach it from $(0, 0)$.

Work space

Practice 3: Plotting points

Name: _____ Date: _____

1. Plot $D(1, 1)$, $E(1, 5)$, $F(5, 5)$, and $G(5, 1)$ on a coordinate grid. Connect them in order, then connect G back to D . What shape is formed?

Work space

2. Plot $H(2, 6)$, $I(4, 6)$, and $J(4, 3)$. Which points are tied for highest? Which two have the same x-coordinate?

Work space

Practice 4: Horizontal and vertical movement

Name: _____ Date: _____

1. Start at $K(2, 3)$. Move 4 units right. Write the new ordered pair and explain which coordinate changed.

Work space

2. Start at $L(6, 5)$. Move 3 units down. Write the new ordered pair and explain why.

Work space

Practice 5: Translating points

Name: _____ Date: _____

Use capital letters for points and parentheses for coordinates. A prime mark, as in M' , names a point after a transformation.

1. Translate $M(1, 4)$ 3 units right and 2 units up. Give M' and show the coordinate changes.

Work space

2. A point $N(7, 2)$ is translated 5 units left and 1 unit down. Give N' .

Work space

Practice 6: Translating figures

Name: _____ Date: _____

1. Triangle PQR has vertices $P(1, 1)$, $Q(3, 1)$, and $R(2, 4)$. Translate it 2 units right. List P' , Q' , and R' .

Work space

2. Rectangle $STUV$ has vertices $(2, 2)$, $(6, 2)$, $(6, 4)$, $(2, 4)$. Translate it 1 unit left and 3 units up. List all four new vertices.

Work space

Practice 7: Reflections across a horizontal line

Name: _____ Date: _____

Use the horizontal mirror line through 4 on the y-axis (the line $y = 4$). A reflected point is the same distance from the mirror line on the opposite side.

1. Reflect $A(3, 5)$ across the line $y = 4$. Give A' and describe how its distance from the mirror line helps you.

Work space

2. Reflect triangle vertices $B(1, 2)$, $C(4, 2)$, and $D(2, 6)$ across the line $y = 4$. List the image coordinates.

Work space

Practice 8: Reflections across a vertical line

Name: _____ Date: _____

Use the vertical mirror line through 4 on the x-axis (the line $x = 4$).

1. Reflect $E(6, 2)$ across the line $x = 4$. Give E' and state which coordinate changes.

Work space

2. Reflect rectangle vertices $(1, 1)$, $(4, 1)$, $(4, 3)$, $(1, 3)$ across the line $x = 4$. List the four image vertices.

Work space

Practice 9: Reflection reasoning

Name: _____ Date: _____

1. Point F is $(4, 4)$. What happens when it is reflected across the vertical line $x = 4$? Explain.

Work space

2. A point and its reflection across the horizontal line $y = 4$ are $(5, 3)$ and $(5, \square)$. Fill the blank and explain.

Work space

Practice 10: Quarter-turn rotations

Name: _____ Date: _____

1. A toy arrow points up. After a quarter-turn clockwise, which direction does it point? After another quarter-turn clockwise?

Work space

2. Square $ABCD$ has vertices $A(1, 1)$, $B(3, 1)$, $C(3, 3)$, and $D(1, 3)$. Sketch a quarter-turn clockwise around its center $(2, 2)$ and label the image vertices A' , B' , C' , and D' .

Work space

Practice 11: Rotation around a point

Name: _____ Date: _____

1. Point G is 2 units right of center $O(4, 4)$. After a quarter-turn clockwise around O , where is G' ?

Work space

2. Point H is 3 units above center $O(5, 5)$. After a quarter-turn counterclockwise around O , where is H' ?

Work space

Practice 12: Distance on a grid

Name: _____ Date: _____

1. Find the horizontal distance between $J(2, 4)$ and $K(7, 4)$.

Work space

2. Find the vertical distance between $L(3, 1)$ and $M(3, 6)$.

Work space

Practice 13: Perimeter from coordinates

Name: _____ Date: _____

1. A rectangle has vertices $(1, 1)$, $(6, 1)$, $(6, 4)$, $(1, 4)$. Find its side lengths and perimeter.

Work space

2. A square has vertices $(2, 2)$, $(5, 2)$, $(5, 5)$, $(2, 5)$. Find its perimeter. Explain how the coordinates help.

Work space

Practice 14: Area from coordinates

Name: _____ Date: _____

1. Find the area of the rectangle with vertices $(2, 1)$, $(8, 1)$, $(8, 4)$, $(2, 4)$.

Work space

2. A square runs from $x = 3$ to $x = 7$ and $y = 2$ to $y = 6$. Find its side length and area.

Work space

Practice 15: Congruent figures

Name: _____ Date: _____

1. Two rectangles have side lengths 4 units and 7 units. Are they congruent? What does congruent mean?

Work space

2. A triangle is translated 5 units right. Does its size or shape change? Explain why the original and image are congruent.

Work space

Practice 16: Comparing transformations

Name: _____ Date: _____

1. Which transformation slides a figure without turning or flipping it? Describe it.

Work space

2. A point at $(4, 2)$ is reflected to $(6, 2)$. Name the vertical line of reflection and explain how you found it.

Work space

Practice 17: Multi-step transformations

Name: _____ Date: _____

1. Start with $A(2, 3)$. Translate 2 units right, then reflect across the vertical line $x = 5$. Give the final coordinate.

Work space

2. Start with $B(5, 1)$. Reflect across the horizontal line $y = 3$, then translate 1 unit left and 2 units up. Give the final coordinate.

Work space

Practice 18: Coordinate word problems

Name: _____ Date: _____

1. On a map, the library is $(2, 5)$ and the park is $(2, 1)$. How far apart are they if each grid unit is 1 block?

Work space

2. On a grid where east is right and north is up, a robot at $(3, 4)$ moves 4 units east and 3 units south. Where does it stop?

Work space

Practice 19: Design on a coordinate grid

Name: _____ Date: _____

1. Create a rectangle with width 3 and height 2 whose lower-left vertex is $(1, 1)$. List all vertices.

Work space

2. Create a triangle with vertices $(1, 2)$, $(4, 2)$, $(1, 5)$. Find its horizontal and vertical side lengths.

Work space

Practice 20: Mixed review

Name: _____ Date: _____

1. Rectangle $WXYZ$ has vertices $(1, 2)$, $(5, 2)$, $(5, 5)$, $(1, 5)$. Translate it 2 units right, then reflect it across the horizontal line $y = 4$. List the final vertices.

Work space

2. Explain one similarity and one difference between a translation and a reflection. Give a coordinate example for each.

Work space